Automorphism Ratios of Finite Groups

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Abstract

For a finite group G, let $\alpha(G) = \frac{|\operatorname{Aut}(G)|}{|G|}$. We prove that the image of α on finite groups is exactly $\mathbb{Q}^+$. That is, for every $r \in \mathbb{Q}^+$, there exists a finite group G such that $\alpha(G) = r$, thereby answering Problem 21.97 of the Kourovka Notebook in the affirmative. The proof is constructive, combining prime-avoiding metacyclic numerator blocks with perfect denominator blocks built from universal central extensions and Heisenberg-symplectic semidirect products.

Contents

1 Introduction

Let $\mathbb{Q}^+$ denote the positive rationals, and for a finite group G, let

$$\alpha(G) = \frac{|\operatorname{Aut}(G)|}{|G|}.$$

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Problem 21.97 of the Kourovka Notebook, attributed to M. Tărnăuceanu, asks whether every positive rational number r occurs in the form of $\alpha(G)$ for some finite group G [KM26]. Equivalently, the problem asks whether the image of α on finite groups is all of $\mathbb{Q}^+$. Some restricted and associated problems have previously been studied. In the finite abelian case, Tărnăuceanu proved that the set of values $\alpha(A)$, with A finite abelian, is dense in $[0, \infty)$ [Tăr26]. McCulloch subsequently proved that finite abelian groups do not realize all positive rational values (if $r = a/b$ is in lowest terms and $r = \alpha(A)$ for a finite abelian group A, then b is squarefree). He also proved that no odd prime occurs as such a value [McC26]. The structural description of automorphisms of finite abelian groups dates back at least to Ranum [Ran07]. A modern account is given by Hillar and Rhea [HR07]. There are also asymptotic results in the nonabelian p-group direction. González-Sánchez and Jaikin-Zapirain constructed, for each prime p, finite p-groups G_i such that $|\operatorname{Aut}(G_i)|/|G_i| \to 0$ [GJ15]. Such results show that the ratio can be made very small, but they do not by themselves give exact realization of arbitrary rational values.

We prove that the image of α on finite groups is exactly $\mathbb{Q}^+$.

Theorem 1.1. *For every $r \in \mathbb{Q}^+$, there exists a finite group G such that $\alpha(G) = r$.*

The first observation is the elementary identity

$$\alpha(G) = \frac{|\operatorname{Out}(G)|}{|Z(G)|}.$$

Consequently, the problem is not solely about the size of $\operatorname{Aut}(G)$, but also about constructing finite groups such that the center and the outer automorphism group have a particular ratio.

To this end, considering the numerator is relatively simple. For every positive integer d, if $q \equiv 1 \pmod{d}$ is a suitable prime and $m = (q-1)/d$, then a metacyclic group $C_q \rtimes C_m$ has $\alpha = d$. However, we need a strengthened form with prime avoidance. Following the parity condition used below, the order of the metacyclic group can be made coprime to a prescribed finite set of primes. This follows from the Chinese remainder theorem and Dirichlet's theorem on primes in arithmetic progressions [Apo76]. In the final process, we apply this prime-avoiding construction with $d = 2a$, which allows us to avoid the prime 2 as well as all primes dividing the denominator block.

By comparison, the main difficulty is denominator control. We first construct groups with $\alpha = 2^{-e}$ for all $e \geq 0$. To do this, we use universal central extensions, Schur multipliers, and a finite-code device. If Q is finite, perfect, and centerless, $E \to Q$ is its universal central extension with kernel M, and $N \leq M$, then $G = E/N$ satisfies that

$$\alpha(G) = \frac{|\operatorname{Stab}_{\operatorname{Out}(Q)}(N)|}{|M/N|},$$

where the stabilizer is setwise and $\operatorname{Out}(Q)$ acts on M through the unique lifts of automorphisms of Q to E. We prove this stabilizer formula from the universal property of the universal central extension, after recalling the standard identification of its kernel with the homological Schur multiplier [Kar87; Bro82]. For the main dyadic construction, we take $Q = B^k$, where B is the Baby Monster. Then $\operatorname{Out}(B^k) \cong S_k$, and a binary-code construction chooses $N \leq M \cong \mathbb{F}_2^k$ so that no nontrivial coordinate permutation stabilizes N. This handles all dyadic exponents $e \geq 9$, while two explicit asymmetric binary subsets handle the exponents 5 and 6. The exponent $e = 0$ is handled by the trivial group. The small exponents $1 \leq e \leq 4$ are supplied by standard ATLAS data for the Baby Monster, the Rudvalis group, and $PSL_3(4)$, and the cases $e = 7$ and $e = 8$ are obtained by multiplying the explicit code blocks by the universal central extension of $PSL_3(4)$ [Con+85; Wil09; Wil+a; Wil+c; Wil+b; BMO17].

The odd-prime construction is the primary new mechanism. Let p be odd, let (V, ω) be a $2n$-dimensional symplectic vector space over $\mathbb{F}_p$, and let $H(V)$ be the associated finite Heisenberg group. For $n \geq 3$, define

$$J_{p,n} = H(V) \rtimes \mathrm{Sp}(V, \omega).$$

We prove directly that $J_{p,n}$ is perfect, that $Z(J_{p,n}) \cong C_p$, and that, indeed, every outer automorphism class has a representative induced by a symplectic similitude. Consequently $\mathrm{Out}(J_{p,n}) \cong \mathbb{F}_p^\times$, and hence $\alpha(J_{p,n}) = (p-1)/p$. The finite symplectic group facts needed here are standard, as follows. For odd p and $n \geq 3$, the group $\mathrm{Sp}_{2n}(p)$ is perfect, has center $\{\pm I\}$, has no nontrivial normal p-subgroup, and acts irreducibly on its natural module [Gro02; Tay92; KL90].

The factor $p-1$ in $(p-1)/p$ is not an obstruction. Since every prime divisor of $p-1$ is smaller than p, one can recursively cancel it using denominator blocks for smaller primes. This descending prime recursion gives a finite perfect group D_b with $\alpha(D_b) = 1/b$ for every positive integer b. Finally, for $r = a/b$, we multiply D_{2b} by a prime-avoiding metacyclic numerator block with ratio $2a$, obtaining ratio a/b. The internal direct-product steps in the denominator construction are justified by some characteristic-factor arguments. The dyadic factor has a centerless quotient that is a direct product of nonabelian simple groups, while each odd-prime Jacobi quotient has a unique abelian minimal normal subgroup whose order is chosen to distinguish it from the others. The final numerator-denominator product has coprime factors by construction. Together, these constructions realize every element of $\mathbb{Q}^+$, and no other values are possible since $\alpha(G)$ is rational for every finite group G.

2 Preliminaries

2.1 A simple ratio identity

Throughout this paper, all groups are finite. For a group G, we denote by $Z(G)$ its center, $\mathrm{Inn}(G) \trianglelefteq \mathrm{Aut}(G)$ for its inner automorphism group, and $\mathrm{Out}(G) = \mathrm{Aut}(G)/\mathrm{Inn}(G)$ for its outer automorphism group. We then have the following elementary identity.

Lemma 2.1. *For every finite group G, one has*

$$\frac{|\mathrm{Aut}(G)|}{|G|} = \frac{|\mathrm{Out}(G)|}{|Z(G)|}.$$

Proof. Let $c_g \colon G \to G$ denote conjugation by g, $c_g(x) = gxg^{-1}$. Then, the map $G \to \mathrm{Inn}(G)$, with $g \mapsto c_g$ is a surjective homomorphism with kernel $Z(G)$. Hence $|\mathrm{Inn}(G)| = |G|/|Z(G)|$. Since $\mathrm{Inn}(G) \trianglelefteq \mathrm{Aut}(G)$ and $\mathrm{Out}(G) = \mathrm{Aut}(G)/\mathrm{Inn}(G)$, we have $|\mathrm{Aut}(G)| = |\mathrm{Inn}(G)| \cdot |\mathrm{Out}(G)|$. Therefore,

$$\frac{|\mathrm{Aut}(G)|}{|G|} = \frac{(|G|/|Z(G)|)\,|\mathrm{Out}(G)|}{|G|} = \frac{|\mathrm{Out}(G)|}{|Z(G)|}.$$

□

Notation 2.2. Recall that $\alpha(G) = |\mathrm{Aut}(G)|/|G|$. Thus, by Lemma 2.1, $\alpha(G) = |\mathrm{Out}(G)|/|Z(G)|$.

2.2 Universal central extensions and stabilizers

We recall a small amount of Schur multiplier theory that is used in the proof.

A central extension of a group Q is a surjection $\pi\colon E \to Q$ with $\ker(\pi) \le Z(E)$. Although all groups we discuss later are finite, universal central extensions are understood in the usual category of groups. If Q is perfect, a central extension $\pi\colon E \to Q$ is called universal if, for every central extension $\rho\colon X \to Q$, there is a unique homomorphism $f\colon E \to X$ such that $\rho \circ f = \pi$.

We use the homological convention for the Schur multiplier, $M(Q) = H_2(Q, \mathbb{Z})$. With this convention, the kernel of the universal central extension of a perfect group Q is naturally isomorphic to $M(Q)$. Since Q is finite in all applications below, the group $H_2(Q, \mathbb{Z})$ is finite, and hence the universal central extension of such a Q is again finite. With respect to the cohomological convention, for finite Q with trivial Q-action on $\mathbb{C}^\times$, the universal coefficient theorem gives that

$$H^2(Q, \mathbb{C}^\times) \cong \mathrm{Hom}(H_2(Q, \mathbb{Z}), \mathbb{C}^\times),$$

since $\mathbb{C}^\times$ is divisible. Below, we use only the universal property, the identification of the kernel with the homological Schur multiplier, and the elementary consequences noted in the next fact. References on this subject include [Kar87; Bro82].

Fact 2.3. *Let Q be a finite perfect group, and let $1 \to M \to E \xrightarrow{\pi} Q \to 1$ be its universal central extension. Then E is perfect and $M \le Z(E)$. If Q is centerless, then $Z(E) = M$. Moreover, every automorphism of Q lifts uniquely to an automorphism of E. The resulting action of* $\mathrm{Aut}(Q)$ *on M factors through* $\mathrm{Out}(Q)$.

Proof. Note that the inclusion $M \le Z(E)$ is part of the definition of a central extension. Thus, we first prove that E is perfect. Since Q is perfect and π is surjective, $\pi(E') = Q' = Q$. Hence, $\pi|_{E'}\colon E' \to Q$ is surjective. Then, its kernel is $E' \cap M$, which is central in E', so $\pi|_{E'}\colon E' \to Q$ is a central extension. So by universality, there is a homomorphism $f\colon E \to E'$ such that $(\pi|_{E'}) \circ f = \pi$. If $i\colon E' \hookrightarrow E$ denotes inclusion, then $i \circ f\colon E \to E$ and $\mathrm{id}_E\colon E \to E$ are both homomorphisms over Q. By uniqueness in the universal property, $i \circ f = \mathrm{id}_E$. Therefore every element of E lies in E', and $E = E'$.

Now, suppose that Q is centerless. If $x \in Z(E)$, then $\pi(x) \in Z(Q) = 1$, so $x \in M$. Since $M \le Z(E)$, this gives $Z(E) = M$.

Let $\alpha \in \mathrm{Aut}(Q)$. Then the map $\alpha^{-1} \circ \pi\colon E \to Q$ is again a central extension, with kernel again M. So by universality, there is a unique homomorphism $\widetilde{\alpha}\colon E \to E$ such that $(\alpha^{-1} \circ \pi) \circ \widetilde{\alpha} = \pi$, or equivalently $\pi \circ \widetilde{\alpha} = \alpha \circ \pi$. Applying this same construction to α^{-1} yields a homomorphism $\widetilde{\alpha^{-1}}\colon E \to E$ with $\pi \circ \widetilde{\alpha^{-1}} = \alpha^{-1} \circ \pi$. Then $\widetilde{\alpha^{-1}} \circ \widetilde{\alpha}$ and id_E are both homomorphisms $E \to E$ over Q, so uniqueness gives $\widetilde{\alpha^{-1}} \circ \widetilde{\alpha} = \mathrm{id}_E$. Similarly, $\widetilde{\alpha} \circ \widetilde{\alpha^{-1}} = \mathrm{id}_E$, and thus, $\widetilde{\alpha} \in \mathrm{Aut}(E)$.

If $\theta\colon E \to E$ is another lift of α, then $(\alpha^{-1} \circ \pi) \circ \theta = \pi = (\alpha^{-1} \circ \pi) \circ \widetilde{\alpha}$. By uniqueness for the central extension $\alpha^{-1} \circ \pi\colon E \to Q$, one has $\theta = \widetilde{\alpha}$.

Since $\pi \circ \widetilde{\alpha} = \alpha \circ \pi$, the automorphism $\widetilde{\alpha}$ maps $M = \ker \pi$ onto itself. If $\alpha, \beta \in \mathrm{Aut}(Q)$, then $\widetilde{\alpha} \circ \widetilde{\beta}$ and $\widetilde{\alpha\beta}$ are both lifts of $\alpha\beta$, and hence $\widetilde{\alpha} \circ \widetilde{\beta} = \widetilde{\alpha\beta}$. Thus, restriction to M defines an action of $\mathrm{Aut}(Q)$ on M.

Finally, let $\alpha = c_q$ be inner, and choose $e \in E$ with $\pi(e) = q$. Conjugation by e is a lift of c_q, so by uniqueness it is the unique lift. Since $M \leq Z(E)$, conjugation by e fixes M pointwise. Therefore $\mathrm{Inn}(Q)$ acts trivially on M, and the action of $\mathrm{Aut}(Q)$ on M factors through $\mathrm{Out}(Q)$. □

Lemma 2.4. *Let Q be finite, perfect, and centerless. Let $1 \to M \to E \xrightarrow{\pi} Q \to 1$ be the universal central extension of Q, so that $M = Z(E)$. Let $N \leq M$, and let $G = E/N$. Then $Z(G) = M/N$, and*

$$\mathrm{Out}(G) \cong \mathrm{Stab}_{\mathrm{Out}(Q)}(N),$$

where Stab denotes setwise stabilizer and $\mathrm{Out}(Q)$ acts on M through the unique lifts of automorphisms of Q to E. Consequently,

$$\alpha(G) = \frac{|\,\mathrm{Stab}_{\mathrm{Out}(Q)}(N)|}{|M/N|}.$$

Proof. Let $\nu\colon E \to G = E/N$ be the quotient map, and let $\bar{\pi}\colon G \to Q$ be the induced map. Thus $\pi = \bar{\pi} \circ \nu$, and $\ker(\bar{\pi}) = M/N$.

First we compute the center. If $xN \in Z(E/N)$, then $[x, e] \in N \leq M$ for every $e \in E$. Hence $xM \in Z(E/M)$. Since $E/M \cong Q$ and Q is centerless, this gives $x \in M$. Thus $Z(G) \leq M/N$. The reverse inclusion follows from $M \leq Z(E)$. Hence $Z(G) = M/N$, and therefore $G/Z(G) \cong Q$.

Since E is perfect by Fact 2.3, its quotient $G = E/N$ is also perfect. Every automorphism of G preserves $Z(G)$, and hence induces an automorphism of $G/Z(G) \cong Q$. We proceed to show that the resulting natural map $\mathrm{Aut}(G) \to \mathrm{Aut}(Q)$ is injective. Let $\varphi \in \mathrm{Aut}(G)$ induce the identity on $G/Z(G)$. Then for each $g \in G$, there is some $z_g \in Z(G)$ such that $\varphi(g) = gz_g$. Therefore $\varphi([g, h]) = [g, h]$ for all $g, h \in G$, because central factors do not affect commutators. Since G is perfect, $G = G'$, and hence $\varphi = \mathrm{id}_G$. Thus $\mathrm{Aut}(G) \to \mathrm{Aut}(Q)$ is injective.

We now identify its image. Let $A \leq \mathrm{Aut}(Q)$ be the subgroup consisting of those $\beta \in \mathrm{Aut}(Q)$ whose unique lift $\widetilde{\beta} \in \mathrm{Aut}(E)$ stabilizes N. If $\beta \in A$, then $\widetilde{\beta}$ descends to an automorphism of $G = E/N$, and this descended automorphism induces β on Q. Thus A is contained in the image of $\mathrm{Aut}(G) \to \mathrm{Aut}(Q)$.

Conversely, let $\varphi \in \mathrm{Aut}(G)$, and let $\beta \in \mathrm{Aut}(Q)$ be the automorphism induced by φ. Let $\widetilde{\beta} \in \mathrm{Aut}(E)$ be the unique lift of β, so that $\pi \circ \widetilde{\beta} = \beta \circ \pi$. The map $\beta^{-1} \circ \bar{\pi}\colon G \to Q$ is a central extension, since its kernel is $M/N = Z(G)$. The two homomorphisms $\varphi \circ \nu\colon E \to G$ and $\nu \circ \widetilde{\beta}\colon E \to G$ both lie over Q with respect to this central extension, since

$$(\beta^{-1} \circ \bar{\pi}) \circ \varphi \circ \nu = \pi \quad \text{and} \quad (\beta^{-1} \circ \bar{\pi}) \circ \nu \circ \widetilde{\beta} = \pi.$$

By the universal property of E, these two homomorphisms are equal. Hence $\varphi \circ \nu = \nu \circ \widetilde{\beta}$. In particular, if $n \in N = \ker(\nu)$, then $\nu(\widetilde{\beta}(n)) = \varphi(\nu(n)) = 1$, so $\widetilde{\beta}(n) \in N$. Thus $\widetilde{\beta}(N) \leq N$. Applying the same argument to φ^{-1}, whose induced automorphism of Q is β^{-1} and whose unique lift to E is $\widetilde{\beta}^{-1}$, gives $\widetilde{\beta}^{-1}(N) \leq N$. Therefore $\widetilde{\beta}(N) = N$, and so $\beta \in A$.

Therefore $\mathrm{Aut}(G) \cong A = \mathrm{Stab}_{\mathrm{Aut}(Q)}(N)$, where the stabilizer is taken with respect to the action on M through unique lifts. Inner automorphisms of Q act trivially on M, by Fact 2.3, and hence lie in A. Under the identification $G/Z(G) \cong Q$, conjugation by $g \in G$ maps to conjugation by $\bar{\pi}(g) \in Q$. Since $\bar{\pi}\colon G \to Q$ is surjective, the image of $\mathrm{Inn}(G)$ is all of $\mathrm{Inn}(Q)$. Moreover, if conjugation by $g \in G$

induces the identity on Q, then $\bar{\pi}(g) \in Z(Q) = 1$, so $g \in \ker(\bar{\pi}) = Z(G)$, and hence conjugation by g is trivial. Thus $\operatorname{Inn}(G)$ identifies with $\operatorname{Inn}(Q) \leq A$. Passing to quotients gives

$$\operatorname{Out}(G) \cong A/\operatorname{Inn}(Q) \cong \operatorname{Stab}_{\operatorname{Out}(Q)}(N).$$

Finally, by Lemma 2.1 and $Z(G) = M/N$, we obtain

$$\alpha(G) = \frac{|\operatorname{Out}(G)|}{|Z(G)|} = \frac{|\operatorname{Stab}_{\operatorname{Out}(Q)}(N)|}{|M/N|}.$$

□

Corollary 2.5. *Let Q be finite, perfect, and centerless, and let $1 \to M \to E \to Q \to 1$ be the universal central extension. Then*

$$\alpha(E) = \frac{|\operatorname{Out}(Q)|}{|M|}.$$

Proof. This is Lemma 2.4 with $N = 1$. □

2.3 Direct products with separated centerless quotients

We shall repeatedly multiply groups whose orders are not necessarily coprime. The following elementary criterion is the substitute for coprime-order multiplicativity.

Lemma 2.6. *Let $G = G_1 \times \cdots \times G_t$, where each G_i is perfect and $G_i/Z(G_i)$ is centerless. Let $Q_i = G_i/Z(G_i)$. Suppose that every automorphism of $Q = Q_1 \times \cdots \times Q_t$ preserves each direct factor Q_i. Then every automorphism of G preserves each direct factor G_i. Consequently $\operatorname{Aut}(G) \cong \operatorname{Aut}(G_1) \times \cdots \times \operatorname{Aut}(G_t)$, and $\alpha(G) = \prod_i \alpha(G_i)$.*

Proof. The center of G is $\prod_i Z(G_i)$, so $G/Z(G) \cong Q$. Let $\varphi \in \operatorname{Aut}(G)$, and let $\bar{\varphi}$ be the induced automorphism of $G/Z(G) \cong Q$. By the separation hypothesis, $\bar{\varphi}$ preserves each direct factor Q_i. Let P_i be the preimage of Q_i under the quotient map $G \to G/Z(G) \cong Q$. Then $P_i = G_i \times \prod_{j \neq i} Z(G_j)$, and $\varphi(P_i) = P_i$. Since G_i is perfect and the remaining factors in P_i are central, one has $P_i' = G_i$. Therefore

$$\varphi(G_i) = \varphi(P_i') = \varphi(P_i)' = P_i' = G_i.$$

Thus every automorphism of G preserves every factor G_i. Restriction to the factors gives $\operatorname{Aut}(G) \cong \operatorname{Aut}(G_1) \times \cdots \times \operatorname{Aut}(G_t)$, and hence $\alpha(G) = \prod_i \alpha(G_i)$. □

Lemma 2.7. *If A and B are finite groups with $\gcd(|A|, |B|) = 1$, then $\operatorname{Aut}(A \times B) \cong \operatorname{Aut}(A) \times \operatorname{Aut}(B)$, and $\alpha(A \times B) = \alpha(A)\alpha(B)$.*

Proof. Let π_A be the set of primes dividing $|A|$. We identify A with $A \times 1 \leq A \times B$. If $(a, b) \in A \times B$, then $|(a, b)| = \operatorname{lcm}(|a|, |b|)$. Since $\gcd(|A|, |B|) = 1$, the element (a, b) has π_A-number order if and only if $b = 1$. Thus $A \times 1$ is precisely the set of elements of $A \times B$ whose orders have all prime divisors in π_A. This description is invariant under automorphisms, so $A \times 1$ is characteristic. The same argument shows that $1 \times B$ is characteristic.

Therefore every automorphism of $A \times B$ preserves both direct factors. Restriction to the two factors then yields $\mathrm{Aut}(A \times B) \cong \mathrm{Aut}(A) \times \mathrm{Aut}(B)$. Hence,

$$\alpha(A \times B) = \frac{|\operatorname{Aut}(A)|\,|\operatorname{Aut}(B)|}{|A|\,|B|} = \alpha(A)\alpha(B).$$

□

2.4 Finite symplectic groups

We shall use a few standard facts regarding finite symplectic groups. Some references on the subject include [Gro02; Tay92; KL90].

Fact 2.8. *Let p be an odd prime, let $n \geq 3$, and let V be a $2n$-dimensional vector space over $\mathbb{F}_p$ equipped with a nondegenerate alternating bilinear form ω. Let $S = \mathrm{Sp}(V, \omega)$. Then S is perfect, $Z(S) = \{\pm I\}$, S has no nontrivial normal p-subgroup, and the natural S-module V is irreducible. Also the natural action is faithful, equivalently $C_S(V) = 1$.*

Proof. We recall the standard arguments and references. It is standard that $S' = S$ in this range [Tay92]. Thus S is perfect.

Now, we compute the center. For $0 \neq u \in V$, define $\tau_u \colon V \to V$ by $\tau_u(v) = v + \omega(v, u)u$. Since ω is alternating, a direct calculation gives $\omega(\tau_u(v), \tau_u(w)) = \omega(v, w)$ for all $v, w \in V$, so $\tau_u \in S$. If $g \in Z(S)$, then g commutes with every τ_u. Since $g\tau_u g^{-1} = \tau_{gu}$, centrality gives $\tau_{gu} = \tau_u$ for every $u \neq 0$. But $\mathrm{im}(\tau_u - \mathrm{id}) = \langle u \rangle$, so $\langle gu \rangle = \langle u \rangle$ for every $u \neq 0$. Hence g preserves every one-dimensional subspace of V, and therefore $g = \lambda I$ for some $\lambda \in \mathbb{F}_p^{\times}$. The symplectic condition gives $\lambda^2 = 1$, so $g = \pm I$. Conversely, $\pm I \in S$ and both scalar maps commute with every element of S. Thus $Z(S) = \{\pm I\}$.

By the standard simplicity theorem for finite symplectic groups, $S/Z(S) \cong PSp_{2n}(p)$ is nonabelian simple for $n \geq 3$. Now let $P \trianglelefteq S$ be a normal p-subgroup. Its image in $S/Z(S)$ is a normal p-subgroup. Since $S/Z(S)$ is nonabelian simple, this image is either trivial or all of $S/Z(S)$. It cannot be all of $S/Z(S)$, since $S/Z(S)$ is not a p-group. Hence the image is trivial, so $P \leq Z(S) = \{\pm I\}$. Since p is odd, this forces $P = 1$.

Finally, the natural module V is irreducible. Indeed, let $0 \neq W \leq V$ be S-invariant, and choose $0 \neq w \in W$. If $u \in V$ satisfies $\omega(w, u) \neq 0$, then $\tau_u(w) - w = \omega(w, u)u \in W$, so $u \in W$. Thus W contains every vector outside $w^{\perp}$. If $y \in w^{\perp}$, choose $u \notin w^{\perp}$. Then, $u + y \notin w^{\perp}$, so both u and $u + y$ lie in W, and hence $y \in W$. Therefore $W = V$. The equality $C_S(V) = 1$ follows because S is realized as a subgroup of $\mathrm{GL}(V)$, so the natural action is faithful. □

2.5 Finite Heisenberg groups

Let p be an odd prime, and let (V, ω) be a nonzero finite-dimensional symplectic vector space over $\mathbb{F}_p$. The associated finite Heisenberg group is $H(V) = V \times \mathbb{F}_p$, with multiplication

$$(v, a)(w, b) = \left(v + w,\ a + b + \frac{1}{2}\omega(v, w)\right).$$

Here $1/2$ denotes the inverse of 2 in $\mathbb{F}_p$. Associativity follows from bilinearity of ω, as both $((u,a)(v,b))(w,c)$ and $(u,a)((v,b)(w,c))$ have first coordinate $u+v+w$ and second coordinate $a+b+c+\frac{1}{2}\omega(u,v)+\frac{1}{2}\omega(u,w)+\frac{1}{2}\omega(v,w)$.

Lemma 2.9. *Let $H = H(V)$. Then H has exponent p, $Z(H) = H' = \{0\} \times \mathbb{F}_p$, and, with the convention $[x,y] = xyx^{-1}y^{-1}$, one has*

$$[(v,0),(w,0)] = (0,\omega(v,w))$$

for all $v, w \in V$. Moreover, if an automorphism of H acts trivially on $H/Z(H)$ and on $Z(H)$, then it is inner.

Proof. The identity element is $(0,0)$, and a direct calculation gives $(v,a)^{-1} = (-v,-a)$. With the commutator convention above, another direct calculation gives

$$[(v,a),(w,b)] = (0,\omega(v,w)).$$

Hence $H' \leq \{0\} \times \mathbb{F}_p$, and $\{0\} \times \mathbb{F}_p$ is central. Since $V \neq 0$ and ω is nondegenerate, there exist $v, w \in V$ with $\omega(v,w) = 1$. Thus $[(cv,0),(w,0)] = (0,c)$ for every $c \in \mathbb{F}_p$, and so $H' = \{0\} \times \mathbb{F}_p$.

If $(v,a) \in Z(H)$, then $\omega(v,w) = 0$ for every $w \in V$, and hence $v = 0$ by nondegeneracy. Therefore $Z(H) \leq \{0\} \times \mathbb{F}_p$. The reverse inclusion was already observed, so $Z(H) = H' = \{0\} \times \mathbb{F}_p$.

For $m \geq 0$, induction gives $(v,a)^m = (mv, ma)$, because $\omega(v,v) = 0$. Hence every element has order dividing p, and every nonidentity element has order p. Thus H has exponent p.

Now let $\theta \in \mathrm{Aut}(H)$ act trivially on $H/Z(H)$ and on $Z(H)$. For each $v \in V$, there is a unique $\lambda(v) \in \mathbb{F}_p$ such that $\theta(v,0) = (v,\lambda(v))$. Since θ fixes $Z(H)$ pointwise, it follows that $\theta(v,a) = (v, a+\lambda(v))$. Comparing $\theta((v,0)(w,0))$ with $\theta(v,0)\theta(w,0)$ gives $\lambda(v+w) = \lambda(v)+\lambda(w)$, so $\lambda\colon V \to \mathbb{F}_p$ is $\mathbb{F}_p$-linear.

By nondegeneracy of ω, there exists $u \in V$ such that $\lambda(v) = \omega(u,v)$ for every $v \in V$. A direct calculation gives

$$(u,0)(v,a)(u,0)^{-1} = (v, a+\omega(u,v)).$$

Thus θ is conjugation by $(u,0)$, and hence is inner. □

3 Construction of all rational values

3.1 Integer blocks with prime avoidance

We begin by realizing all positive integers in a form that will later allow coprime direct products.

Lemma 3.1. *Let $d \geq 1$, and let Π be a finite set of primes. Assume that $2 \in \Pi$ implies $2 \mid d$. Then there exists a finite group A such that $\alpha(A) = d$ and no prime in Π divides $|A|$.*

Proof. We shall choose a prime q with $q \equiv 1 \pmod{d}$, and then use a metacyclic group $C_q \rtimes C_m$, where $m = (q-1)/d$. The aim of this is to choose q so that both q and m avoid all primes in Π.

Represent d as $\prod_{r \mid d} r^{a_r}$, where $a_r = v_r(d) > 0$. We impose congruences for the primes in $\Pi \cup \{r : r \mid d\}$. If $r \mid d$ and $r \notin \Pi$, impose $q \equiv 1 \pmod{r^{a_r}}$. If $r \mid d$ and $r \in \Pi$, impose $q \equiv 1+d \pmod{r^{a_r+1}}$. Finally, if $r \in \Pi$ and $r \nmid d$, then r is odd by hypothesis, and impose $q \equiv 2 \pmod{r}$. These congruences have pairwise coprime prime-power moduli, so the Chinese remainder theorem gives a residue class modulo an integer M.

The chosen residue class is coprime to M, since it is congruent to 1 modulo every prime $r \mid d$ with $r \notin \Pi$, congruent to $1 + d \equiv 1 \pmod{r}$ modulo every $r \in \Pi$ with $r \mid d$, and congruent to $2 \pmod{r}$ modulo every odd $r \in \Pi$ with $r \nmid d$. By Dirichlet's theorem on primes in arithmetic progressions [Apo76], there are infinitely many primes in this residue class. Choose such a prime $q > d + 1$, and let $m = (q-1)/d$. Then $m > 1$, $m \mid q - 1$, and $m < q$.

By construction $q \equiv 1 \pmod{d}$. If $r \in \Pi$ and $r \mid d$, then the congruence $q \equiv 1 + d \pmod{r^{a_r+1}}$ gives $v_r(q-1) = a_r$, so $r \nmid m$, and also $r \nmid q$. If $r \in \Pi$ and $r \nmid d$, then $q \equiv 2 \pmod{r}$, so $r \nmid q(q-1)$, and hence $r \nmid qm$. Thus no prime in Π divides qm.

Choose an element $t \in \mathbb{F}_q^\times$ of order m. Let $A = C_q \rtimes C_m$, where a generator of C_m acts on C_q by exponentiation by t. Equivalently,

$$A = \langle a, b \mid a^q = b^m = 1,\ bab^{-1} = a^t \rangle,$$

where t is viewed as an exponent modulo q. Thus $|A| = qm$, the action of C_m on C_q is faithful, and $A/\langle a \rangle \cong C_m$. The subgroup $\langle a \rangle$ is the unique Sylow q-subgroup, as the number of Sylow q-subgroups divides m, is congruent to $1 \pmod{q}$, and $m < q$. Hence $\langle a \rangle$ is characteristic.

Let $\varphi \in \mathrm{Aut}(A)$. Since $\langle a \rangle$ is characteristic, $\varphi(a) = a^u$ for some $u \in \mathbb{F}_q^\times$. Modulo $\langle a \rangle$, the element $\varphi(b)$ must generate $A/\langle a \rangle$, so $\varphi(b) = a^v b^j$, with $v \in \mathbb{F}_q$ and $j \in (\mathbb{Z}/m\mathbb{Z})^\times$. Preserving the relation $bab^{-1} = a^t$ gives

$$a^{ut^j} = \varphi(b)\varphi(a)\varphi(b)^{-1} = \varphi(a^t) = a^{ut}.$$

Since $u \neq 0$ and t has order m, this forces $t^j = t$, and hence $j \equiv 1 \pmod{m}$. Thus every automorphism has the form $a \mapsto a^u$, $b \mapsto a^v b$, with $u \in \mathbb{F}_q^\times$ and $v \in \mathbb{F}_q$.

Conversely, every such choice of u and v defines an automorphism. Indeed, the assignment $a \mapsto a^u$, $b \mapsto a^v b$ preserves $a^q = 1$ and the relation $bab^{-1} = a^t$. It also preserves $b^m = 1$, since

$$(a^v b)^m = a^{v(1+t+\cdots+t^{m-1})} b^m = 1$$

and $1+t+\cdots+t^{m-1} = 0$ in $\mathbb{F}_q$, as $m > 1$ and t has order m. Thus the assignment gives an endomorphism of A. Its image contains a, because a^u generates $\langle a \rangle$, and then also contains $b = a^{-v}(a^v b)$. Hence the endomorphism is surjective, and since A is finite, it is an automorphism.

Therefore $|\mathrm{Aut}(A)| = q(q-1)$, while $|A| = qm$. Hence $\alpha(A) = q(q-1)/(qm) = (q-1)/m = d$. By construction, no prime in Π divides $qm = |A|$. $\square$

3.2 Dyadic denominator blocks

We proceed by constructing groups with ratio 2^{-e}. This is the only place where sporadic simple groups enter the proof. We use standard ATLAS data for Schur multipliers and outer automorphism groups, in both the printed and electronic forms, together with Wilson's text on finite simple groups [Con+85; Wil09; Wil+a; Wil+c; Wil+b]. For the group $PSL_3(4)$, the multiplier and outer automorphism structures used below are also recorded explicitly in [BMO17, § 3.3.1].

Fact 3.2. *The Baby Monster B has Schur multiplier $M(B) \cong C_2$ and trivial outer automorphism group. The Rudvalis group Ru has Schur multiplier $M(Ru) \cong C_2$ and trivial outer automorphism group. For $L = PSL_3(4)$, one has*

$$M(L) \cong C_3 \times C_4 \times C_4, \qquad \operatorname{Out}(L) \cong D_{12},$$

where D_{12} denotes the dihedral group of order 12. In particular,

$$|M(B)| = |M(Ru)| = 2, \quad |M(L)| = 48, \quad |\operatorname{Out}(B)| = |\operatorname{Out}(Ru)| = 1, \quad |\operatorname{Out}(L)| = 12.$$

Corollary 3.3. *There exist finite perfect groups $D_{2,e}$ for $0 \le e \le 4$ such that $\alpha(D_{2,e}) = 2^{-e}$, and such that $D_{2,e}/Z(D_{2,e})$ is a direct product of nonabelian simple groups, with the empty product allowed.*

Proof. For $e = 0$, take $D_{2,0} = 1$. Then $\alpha(D_{2,0}) = 1$, and $D_{2,0}/Z(D_{2,0}) = 1$, the empty direct product.

Let $\widehat{B}$ and $\widehat{Ru}$ be the universal central extensions of B and Ru. By Corollary 2.5 and Fact 3.2, $\alpha(\widehat{B}) = \alpha(\widehat{Ru}) = 1/2$. Take $D_{2,1} = \widehat{B}$. Since B and Ru are nonisomorphic nonabelian simple groups, every automorphism of $B \times Ru$ preserves the two simple direct factors. Hence Lemma 2.6 gives $\alpha(\widehat{B} \times \widehat{Ru}) = 1/4$. Take $D_{2,2} = \widehat{B} \times \widehat{Ru}$.

Let $\widetilde{L}$ be the universal central extension of $L = PSL_3(4)$. Again by Corollary 2.5 and Fact 3.2, $\alpha(\widetilde{L}) = 12/48 = 1/4$. Since L, B, and Ru are pairwise nonisomorphic nonabelian simple groups, every automorphism of $L \times B$, and every automorphism of $L \times B \times Ru$, preserves each displayed simple direct factor. Therefore Lemma 2.6 gives $\alpha(\widetilde{L} \times \widehat{B}) = 1/8$ and $\alpha(\widetilde{L} \times \widehat{B} \times \widehat{Ru}) = 1/16$. Take $D_{2,3} = \widetilde{L} \times \widehat{B}$ and $D_{2,4} = \widetilde{L} \times \widehat{B} \times \widehat{Ru}$.

In each case the chosen group is perfect, and its centerless quotient is a direct product of nonabelian simple groups, with the $e = 0$ case interpreted as the empty direct product. $\square$

For larger dyadic powers, let $\widehat{B} \to B$ denote the universal central extension of the Baby Monster. We need to eliminate the symmetric-group automorphisms that appear in powers of $\widehat{B}$. This is done by a binary-code construction.

Lemma 3.4. *Let $e \ge 1$. Suppose there exists a subset $S \subseteq \mathbb{F}_2^e \setminus \{0\}$ which spans $\mathbb{F}_2^e$ and has trivial setwise stabilizer in $\mathrm{GL}_e(2)$. Then there exists a finite perfect group D such that $\alpha(D) = 2^{-e}$ and $D/Z(D) \cong B^k$, where $k = |S|$.*

Proof. Enumerate $S = \{s_1, \dots, s_k\}$, and let $\varphi\colon \mathbb{F}_2^k \to \mathbb{F}_2^e$ be the surjective linear map whose i-th standard basis vector maps to s_i. Let $N = \ker \varphi$. Thus N has codimension e in $\mathbb{F}_2^k$.

Let $E = \widehat{B}^k$, with the natural map $E \to B^k$. This map is the universal central extension of B^k. Indeed, if $\rho\colon X \to B^k$ is any central extension and $B_i \le B^k$ denotes the i-th simple factor, then $X_i = \rho^{-1}(B_i) \to B_i$ is a central extension. By universality of $\widehat{B}$, there is a unique homomorphism $f_i\colon \widehat{B} \to X_i$ over B_i. For $i \ne j$, the commutator $[f_i(\widehat{B}), f_j(\widehat{B})]$ lies in $\ker \rho$, hence is central in X. Since $f_i(\widehat{B})$ is perfect, as a homomorphic image of $\widehat{B}$, the map $x \mapsto [x, y]$ from $f_i(\widehat{B})$ to $\ker \rho$ is trivial for every fixed $y \in f_j(\widehat{B})$. Hence the subgroups $f_i(\widehat{B})$ commute pairwise, so the f_i combine to a homomorphism $E = \widehat{B}^k \to X$ over B^k. Uniqueness follows by restricting to the k factors of E. Thus $E \to B^k$ is universal.

Since B^k is centerless, Fact 2.3 gives $Z(E) = \ker(E \to B^k)$. Thus $M = Z(E) \cong \mathbb{F}_2^k$. We regard N as a subgroup of M, and set $D = E/N$.

We now identify the relevant outer action. The k simple direct factors of B^k are precisely its minimal normal subgroups, so every automorphism of B^k permutes these simple factors. Hence $\mathrm{Aut}(B^k) \cong \mathrm{Aut}(B)^k \rtimes S_k$. Since $\mathrm{Out}(B) = 1$, quotienting by $\mathrm{Inn}(B^k) = \mathrm{Inn}(B)^k$ gives $\mathrm{Out}(B^k) \cong S_k$. Under the action on M coming from unique lifts to E, inner automorphisms of B^k act trivially by Fact 2.3, while permutations of the simple factors lift by permuting the corresponding factors of E. Therefore $\mathrm{Out}(B^k) \cong S_k$ acts on $M \cong \mathbb{F}_2^k$ by coordinate permutations.

We claim that the stabilizer of N in this S_k-action is trivial. If a coordinate permutation $\sigma \in S_k$ stabilizes N, then φ and $\varphi\sigma^{-1}$ have the same kernel. Since both maps are surjective, there exists $A \in \mathrm{GL}_e(2)$ with $A\varphi = \varphi\sigma^{-1}$. Applying this equality to the standard basis vectors shows that A sends the set of columns S to itself. By hypothesis, $A = 1$. Since the columns s_i are distinct, $\sigma = 1$.

Now Lemma 2.4 gives $\mathrm{Out}(D) = 1$ and $Z(D) \cong M/N$, so $|Z(D)| = 2^e$. Therefore $\alpha(D) = 2^{-e}$. Finally, $D/Z(D) \cong E/M \cong B^k$, and D is perfect because it is a quotient of the perfect group E. $\square$

Lemma 3.5. *For every $e \geq 9$, there exists a subset $S \subseteq \mathbb{F}_2^e \setminus \{0\}$ which spans $\mathbb{F}_2^e$ and has trivial setwise stabilizer in $\mathrm{GL}_e(2)$.*

Proof. Let $V = \mathbb{F}_2^e$, and let $V^\# = V \setminus \{0\}$. Choose a random subset $S \subseteq V^\#$ by including each vector independently with probability $1/2$. Let $N = |V^\#| = 2^e - 1$.

Fix a nonidentity element $A \in \mathrm{GL}_e(2)$, and let $o(A)$ be the number of orbits of A on $V^\#$. The subset S is stabilized by A precisely when it is a union of A-orbits, so $\Pr(A(S) = S) = 2^{o(A)-N}$. Let $f(A)$ be the number of nonzero fixed vectors of A. Since $A \neq 1$, its fixed space has dimension at most $e - 1$, and hence $f(A) \leq 2^{e-1} - 1$. All non-fixed orbits have size at least 2, so

$$o(A) \leq f(A) + \frac{N - f(A)}{2} \leq \frac{(2^e - 1) + (2^{e-1} - 1)}{2} = 3 \cdot 2^{e-2} - 1 \implies$$

$$o(A) - N \leq (3 \cdot 2^{e-2} - 1) - (2^e - 1) = -2^{e-2}$$

So, we have that $\Pr(A(S) = S) \leq 2^{-2^{e-2}}$, and since $|\mathrm{GL}_e(2)| < 2^{e^2}$, the union bound gives

$$\Pr(\exists A \neq 1 : \ A(S) = S) \leq |\mathrm{GL}_e(2)|\, 2^{-2^{e-2}} < 2^{e^2 - 2^{e-2}}.$$

For $e \geq 9$, one has $e^2 < 2^{e-2}$, and hence $2^{e^2 - 2^{e-2}} < 1$. Therefore some $S \subseteq V^\#$ has trivial stabilizer. Such an S necessarily spans V. Indeed, suppose S were contained in a proper subspace $W < V$. If $W \neq 0$, choose $0 \neq w \in W$ and $u \in V \setminus W$. The linear map fixing W pointwise and sending u to $u + w$, extended over a complement to $W + \langle u \rangle$, is a nonidentity element of $\mathrm{GL}_e(2)$ stabilizing S. If $W = 0$, then $S = \varnothing$, which is stabilized by every element of $\mathrm{GL}_e(2)$. In either case, the stabilizer of S is nontrivial, a contradiction. Hence S spans V.

$\square$

Lemma 3.6. *There are spanning subsets $S_5 \subseteq \mathbb{F}_2^5 \setminus \{0\}$ and $S_6 \subseteq \mathbb{F}_2^6 \setminus \{0\}$ with trivial setwise stabilizer in $\mathrm{GL}_5(2)$ and $\mathrm{GL}_6(2)$, respectively.*

Proof. We identify vectors in $\mathbb{F}_2^e$ with integers from 0 to $2^e - 1$, using binary expansion. Thus $1, 2, 4, \ldots, 2^{e-1}$ are the standard basis vectors. Let

$$S_5 = \{1, 2, 4, 8, 16, 17, 21, 24, 26, 27, 30, 31\} \subseteq \mathbb{F}_2^5$$

and

$$S_6 = \{1, 2, 4, 6, 7, 8, 16, 24, 26, 32, 40, 44, 47\} \subseteq \mathbb{F}_2^6.$$

Each set contains the standard basis of its ambient vector space, and hence spans.

It remains to prove that the stabilizers are trivial. We use a finite invariant based on additive triples. For a subset $S \subseteq \mathbb{F}_2^e \setminus \{0\}$, first partition S by the integer $d(x) = |\{y \in S : x + y \in S\}|$. Then recursively refine the partition as follows. Once the elements of S have colors, replace the color of x by the pair consisting of its old color and the multiset of ordered color-pairs $(c(y), c(x + y))$, where $y \in S$ and $x + y \in S$. Every linear automorphism preserving S preserves each partition obtained in this refinement process.

For S_5, the refinement gives the following partitions:

stage	partition
0	$\{1, 4, 26\} \mid \{2, 8, 21\} \mid \{16, 17, 24, 27, 30, 31\}$
1	$\{1\} \mid \{2, 21\} \mid \{4\} \mid \{8\} \mid \{16\} \mid \{17\} \mid \{24\} \mid \{26\} \mid \{27, 30\} \mid \{31\}$
2	$\{1\} \mid \{2\} \mid \{4\} \mid \{8\} \mid \{16\} \mid \{17\} \mid \{21\} \mid \{24\} \mid \{26\} \mid \{27\} \mid \{30\} \mid \{31\}$.

For S_6, the refinement gives:

stage	partition
0	$\{1, 16, 26, 32, 44, 47\} \mid \{2, 4, 6, 7, 8, 24\} \mid \{40\}$
1	$\{1, 16, 26\} \mid \{2, 6\} \mid \{4\} \mid \{7, 8\} \mid \{24\} \mid \{32, 44, 47\} \mid \{40\}$
2	$\{1\} \mid \{2\} \mid \{4\} \mid \{6\} \mid \{7\} \mid \{8\} \mid \{16\} \mid \{24\} \mid \{26\} \mid \{32, 47\} \mid \{40\} \mid \{44\}$
3	$\{1\} \mid \{2\} \mid \{4\} \mid \{6\} \mid \{7\} \mid \{8\} \mid \{16\} \mid \{24\} \mid \{26\} \mid \{32\} \mid \{40\} \mid \{44\} \mid \{47\}$.

These tables are finite certificates. Each entry is obtained by direct addition in $\mathbb{F}_2^e$, and the recursive definition ensures that every linear stabilizer preserves every displayed partition. A short independent script which recomputes the refinement from the displayed definitions and verifies the two terminal partitions appears in Appendix A. Since the terminal partition is discrete in both cases, any linear stabilizer fixes every element of S_5 or S_6. Because each set contains a basis, the stabilizer is the identity. □

Proposition 3.7. *For every $e \geq 0$, there exists a finite perfect group $D_{2,e}$ such that $\alpha(D_{2,e}) = 2^{-e}$ and $D_{2,e}/Z(D_{2,e})$ is a direct product of nonabelian simple groups, with the empty product allowed.*

Proof. The cases $0 \leq e \leq 4$ are supplied by Corollary 3.3.

For $e = 5$ and $e = 6$, let $S_5 \subseteq \mathbb{F}_2^5 \setminus \{0\}$ and $S_6 \subseteq \mathbb{F}_2^6 \setminus \{0\}$ be the spanning subsets with trivial setwise stabilizer given by Lemma 3.6. Applying Lemma 3.4 to S_5 gives a finite perfect group $D_{2,5}$ such that

$\alpha(D_{2,5}) = 2^{-5}$ and $D_{2,5}/Z(D_{2,5}) \cong B^{|S_5|}$. Applying the same lemma to S_6 gives a finite perfect group $D_{2,6}$ such that $\alpha(D_{2,6}) = 2^{-6}$ and $D_{2,6}/Z(D_{2,6}) \cong B^{|S_6|}$. Thus the cases $e = 5$ and $e = 6$ are handled, and their centerless quotients are direct products of nonabelian simple groups.

Let $L = PSL_3(4)$, and let $\widetilde{L} \to L$ be its universal central extension. By Corollary 2.5 and Fact 3.2, one has $\alpha(\widetilde{L}) = 12/48 = 1/4 = 2^{-2}$.

We next observe why the relevant direct products have multiplicative α-ratio. Let D be any code block obtained from Lemma 3.4, so that $D/Z(D) \cong B^k$ for some k. The minimal normal subgroups of $B^k \times L$ are the k simple B-factors and the simple L-factor. Since B and L are nonisomorphic nonabelian simple groups, every automorphism of $B^k \times L$ preserves the product B^k of all minimal normal subgroups isomorphic to B, and also preserves the unique minimal normal subgroup isomorphic to L. Hence every automorphism of $B^k \times L$ preserves the two displayed direct factors B^k and L. Therefore Lemma 2.6 applies to $D \times \widetilde{L}$, and gives $\alpha(D \times \widetilde{L}) = \alpha(D)\alpha(\widetilde{L})$.

For $e = 7$, set $D_{2,7} = D_{2,5} \times \widetilde{L}$. Then $\alpha(D_{2,7}) = 2^{-5} \cdot 2^{-2} = 2^{-7}$. For $e = 8$, set $D_{2,8} = D_{2,6} \times \widetilde{L}$. Then $\alpha(D_{2,8}) = 2^{-6} \cdot 2^{-2} = 2^{-8}$. These groups are perfect, and their centerless quotients are respectively $B^{|S_5|} \times L$ and $B^{|S_6|} \times L$, hence direct products of nonabelian simple groups.

Finally, suppose $e \geq 9$. By Lemma 3.5, there exists a spanning subset $S \subseteq \mathbb{F}_2^e \setminus \{0\}$ with trivial setwise stabilizer in $\mathrm{GL}_e(2)$. Applying Lemma 3.4 to this S gives a finite perfect group $D_{2,e}$ such that $\alpha(D_{2,e}) = 2^{-e}$ and $D_{2,e}/Z(D_{2,e}) \cong B^{|S|}$, again a direct product of nonabelian simple groups.

The cases $0 \leq e \leq 4$, $e = 5, 6$, $e = 7, 8$, and $e \geq 9$ together cover all $e \geq 0$, and the proposition follows. □

3.3 Odd-prime Jacobi blocks

We now construct the odd-prime denominator mechanism. Let p be an odd prime, let $n \geq 3$, and let (V, ω) be a $2n$-dimensional symplectic vector space over $\mathbb{F}_p$. Let $H = H(V)$ be the Heisenberg group from Lemma 2.9, and let $S = \mathrm{Sp}(V, \omega)$. The group S acts on H by $s(v, a) = (sv, a)$. This is an action by automorphisms, since $\omega(sv, sw) = \omega(v, w)$ for all $s \in S$ and $v, w \in V$. Define

$$J_{p,n} = H \rtimes S.$$

Lemma 3.8. *For p odd and $n \geq 3$, the group $J_{p,n}$ is perfect and $Z(J_{p,n}) = Z(H) \cong C_p$.*

Proof. Let $J = J_{p,n}$. By Lemma 2.9, one has $H' = Z(H)$. Since S is perfect by Fact 2.8, we have $S = S' \leq J'$. The image of $[H, S]$ in $H/Z(H) \cong V$ is $[V, S]$. Since the natural S-module V is irreducible and nontrivial, $[V, S] \neq 0$, and hence $[V, S] = V$. Therefore the image of $J' \cap H$ in $H/Z(H)$ is all of $H/Z(H)$. Since $Z(H) = H' \leq J'$, it follows that $H \leq J'$. Together with $S \leq J'$, this gives $J' = J$.

Now let $x \in Z(J)$. The image of x in $J/Z(H) \cong V \rtimes S$ is central. We claim that $V \rtimes S$ is centerless. Indeed, if (v, s) centralizes every element of V, then s fixes every vector of V, so $s = 1$ by faithfulness of the natural action. Then $(v, 1)$ centralizes S only if $v \in V^S$. But $V^S = 0$, since V^S is an S-submodule of the irreducible nontrivial S-module V. Hence $V \rtimes S$ is centerless. Thus the image of x in $V \rtimes S$ is trivial, and $x \in Z(H)$. Conversely, $Z(H)$ is central in J, because it is central in H and S acts trivially on $Z(H)$. Therefore $Z(J) = Z(H) \cong C_p$. □

Lemma 3.9. *For p odd and $n \geq 3$, one has $\operatorname{Out}(J_{p,n}) \cong \mathbb{F}_p^\times$.*

Proof. Let $J = J_{p,n} = H \rtimes S$, where $S = \operatorname{Sp}(V, \omega)$, and let $Z = Z(H)$. Let $\operatorname{GSp}(V, \omega)$ denote the group of linear transformations $T \in \operatorname{GL}(V)$ such that $\omega(Tv, Tw) = \mu(T)\omega(v, w)$ for some $\mu(T) \in \mathbb{F}_p^\times$.

First, H is characteristic in J. Indeed, $H = O_p(J)$. Any normal p-subgroup of J maps to a normal p-subgroup of S, and S has no nontrivial normal p-subgroup by Fact 2.8. Thus every automorphism of J preserves H and Z.

Let $\theta \in \operatorname{Aut}(J)$. The induced automorphism of $H/Z \cong V$ is some $T \in \operatorname{GL}(V)$, and the induced automorphism of $Z \cong \mathbb{F}_p$ is multiplication by some $c \in \mathbb{F}_p^\times$. If $v, w \in V$, choose lifts $\widetilde{v}, \widetilde{w} \in H$. Since $[\widetilde{v}, \widetilde{w}]$ corresponds to $\omega(v, w) \in Z$, applying θ gives $\omega(Tv, Tw) = c\,\omega(v, w)$. Hence $T \in \operatorname{GSp}(V, \omega)$, with multiplier c.

The automorphism θ also induces an automorphism β of $J/H \cong S$. The conjugation relation between S and H/Z gives $T(sv) = \beta(s)T(v)$ for all $s \in S$ and $v \in V$. Since S acts faithfully on V, this means $\beta(s) = TsT^{-1}$. Thus T normalizes S. This is automatic for $T \in \operatorname{GSp}(V, \omega)$, because symplectic similitudes normalize the symplectic group.

Conversely, every $T \in \operatorname{GSp}(V, \omega)$, with multiplier $\mu(T)$, defines an automorphism Θ_T of J by

$$\Theta_T(v, a) = (Tv, \mu(T)a) \quad \text{for } (v, a) \in H, \qquad \Theta_T(s) = TsT^{-1} \quad \text{for } s \in S.$$

The formula respects the Heisenberg multiplication because $\omega(Tv, Tw) = \mu(T)\omega(v, w)$, and it respects the semidirect product relation because $TsT^{-1} \in S$.

We now prove that every automorphism of J is a product of an inner automorphism and one of the Θ_T. Given $\theta \in \operatorname{Aut}(J)$, let $T \in \operatorname{GSp}(V, \omega)$ be the similitude just obtained, and replace θ by $\Theta_T^{-1}\theta$. We may therefore assume that θ acts trivially on H/Z, trivially on Z, and trivially on $J/H \cong S$.

The restriction of θ to H is then a central automorphism acting trivially on H/Z and on Z. By Lemma 2.9, this restriction is inner by an element of H. Composing with the inverse of that inner automorphism, we may assume that θ fixes every element of H and still acts trivially on J/H.

For $s \in S$, write $\theta(s) = \delta(s)s$, with $\delta(s) \in H$. Since θ fixes H pointwise, for every $h \in H$ we have

$$shs^{-1} = \theta(shs^{-1}) = \theta(s)h\theta(s)^{-1} = \delta(s)shs^{-1}\delta(s)^{-1}.$$

Thus $\delta(s)$ centralizes $sHs^{-1} = H$, so $\delta(s) \in Z(H)$. Because $Z(H)$ is central in J, the identity $\theta(st) = \theta(s)\theta(t)$ implies that $\delta : S \to Z(H)$ is a homomorphism. Equivalently, after the preceding normalizations, the only possible crossed homomorphism $S \to H$ has image in $Z(H)$, and therefore lies in $\operatorname{Hom}(S, Z(H))$. The group S is perfect and $Z(H)$ is abelian, so $\operatorname{Hom}(S, Z(H)) = 0$. Hence δ is trivial, and $\theta = 1$.

It follows that $\operatorname{Aut}(J)$ is generated by $\operatorname{Inn}(J)$ and the automorphisms Θ_T with $T \in \operatorname{GSp}(V, \omega)$. The intersection of the image of $\operatorname{GSp}(V, \omega)$ with $\operatorname{Inn}(J)$ is exactly the image of $\operatorname{Sp}(V, \omega)$. If $T \in \operatorname{Sp}(V, \omega)$, then Θ_T is conjugation by $T \in S \leq J$. If $\mu(T) \neq 1$, then Θ_T acts nontrivially on $Z(H)$, whereas every inner automorphism fixes $Z(H) = Z(J)$ pointwise. Therefore

$$\operatorname{Out}(J) \cong \operatorname{GSp}(V, \omega)/\operatorname{Sp}(V, \omega).$$

The multiplier map $\mu\colon \mathrm{GSp}(V,\omega) \to \mathbb{F}_p^\times$ has kernel $\mathrm{Sp}(V,\omega)$. It is surjective, as follows. For any $c \in \mathbb{F}_p^\times$, choose a symplectic basis $e_1, \ldots, e_n, f_1, \ldots, f_n$, and define $T_c(e_i) = ce_i$ and $T_c(f_i) = f_i$ for all i. Then $\omega(T_c v, T_c w) = c\,\omega(v,w)$, so $\mu(T_c) = c$. Hence $\mathrm{Out}(J) \cong \mathbb{F}_p^\times$. □

Corollary 3.10. *For every odd prime p and every $n \geq 3$, one has*

$$\alpha(J_{p,n}) = \frac{p-1}{p}.$$

Moreover, $J_{p,n}$ is perfect, $J_{p,n}/Z(J_{p,n})$ is centerless, and $J_{p,n}/Z(J_{p,n})$ has a unique minimal normal subgroup, namely $H/Z(H) \cong V$, of order p^{2n}.

Proof. The formula $\alpha(J_{p,n}) = (p-1)/p$ follows from Lemma 2.1, Lemma 3.8, and Lemma 3.9. The group $J_{p,n}$ is perfect by Lemma 3.8. Since $Z(J_{p,n}) = Z(H)$, one has $J_{p,n}/Z(J_{p,n}) \cong V \rtimes S$, where $S = \mathrm{Sp}(V,\omega)$. This quotient is centerless by the proof of Lemma 3.8.

It remains to prove the assertion about minimal normal subgroups of $V \rtimes S$. The subgroup V is normal, and the normal subgroups of $V \rtimes S$ contained in V are precisely the S-invariant subspaces of V. Since V is irreducible as an S-module, V is a minimal normal subgroup.

Let $N \trianglelefteq V \rtimes S$ be nontrivial. If $N \leq V$, then irreducibility gives $N = V$. Suppose instead that $N \not\leq V$. Then N contains an element (v,s) with $s \neq 1$. Since the natural action of S on V is faithful, there exists $w \in V$ such that $sw \neq w$. The commutator with $(w,1) \in V$ is

$$[(v,s),(w,1)] = (sw - w, 1),$$

which is a nonzero element of $N \cap V$. Thus $N \cap V \neq 0$. Since $N \cap V$ is an S-invariant subspace of V, irreducibility gives $N \cap V = V$, and hence $V \leq N$. Therefore every nontrivial normal subgroup of $V \rtimes S$ contains V, so V is the unique minimal normal subgroup. Finally, $|V| = p^{2n}$. □

3.4 Assembling arbitrary denominator blocks

We now show that every reciprocal integer occurs.

Proposition 3.11. *For every positive integer b, there exists a finite perfect group D_b such that $\alpha(D_b) = 1/b$.*

Proof. Represent b as $\prod_p p^{e_p}$, where only finitely many e_p are nonzero. Let P be the largest prime divisor of b, with the convention that $P = 2$ when $b = 1$. We define nonnegative integers x_p, for primes $p \leq P$, by descending recursion. For each odd prime $p \leq P$, after the x_ℓ with $\ell > p$ have been defined, let

$$x_p = e_p + \sum_{\substack{\ell > p \\ \ell \leq P \\ \ell \text{ prime}}} x_\ell\, v_p(\ell - 1).$$

Finally, let

$$x_2 = e_2 + \sum_{\substack{\ell > 2 \\ \ell \leq P \\ \ell \text{ prime}}} x_\ell\, v_2(\ell - 1).$$

This recursion is finite because the right-hand side for x_p only involves primes larger than p.

For each odd prime $p \leq P$, choose integers $n(p,1), \dots, n(p,x_p)$, all at least 3, such that the orders $p^{2n(p,i)}$ are pairwise distinct among all Jacobi factors. This is possible by choosing the integers $n(p,i)$ distinct for each fixed p, since powers of different primes are automatically distinct. Let

$$D_b = D_{2,x_2} \times \prod_{\substack{p \leq P \\ p \text{ odd}}} \prod_{i=1}^{x_p} J_{p,n(p,i)}.$$

We first justify that the α-ratio of this product is the product of the α-ratios of the displayed factors. Let $Q_0 = D_{2,x_2}/Z(D_{2,x_2})$. Let $I = \{(p,i) : p \leq P,\ p \text{ odd},\ 1 \leq i \leq x_p\}$, and for $a = (p,i) \in I$, let $Q_a = J_{p,n(p,i)}/Z(J_{p,n(p,i)})$. Also let V_a denote the unique minimal normal subgroup of Q_a from Corollary 3.10. Thus $|V_a| = p^{2n(p,i)}$, and these orders are pairwise distinct as a varies.

All products over empty index sets are interpreted as the trivial group. Thus, if $x_2 = 0$, then $D_{2,x_2} = D_{2,0} = 1$ and $Q_0 = 1$. If $I = \varnothing$, then the product over $a \in I$ is 1, and all assertions below involving the subgroups V_a, W_a, and Q_a are vacuous.

Consider the total centerless quotient

$$Q = Q_0 \times \prod_{a \in I} Q_a.$$

By Proposition 3.7, Q_0 is a direct product of nonabelian simple groups, with the empty product allowed. By Corollary 3.10, each $J_{p,n(p,i)}$ is perfect, and hence each quotient $Q_a = J_{p,n(p,i)}/Z(J_{p,n(p,i)})$ is perfect. Also by Corollary 3.10, each Q_a is centerless and has a unique minimal normal subgroup V_a, which is abelian.

We now determine why the minimal normal subgroups of Q come from the displayed direct factors. Let Y be a nontrivial minimal normal subgroup of Q, and let F be one of the direct factors Q_0 or Q_a. Then $[Y,F] \leq Y \cap F$. Since both Y and F are normal in Q, the subgroup $[Y,F]$ is also normal in Q. Thus, if $[Y,F] \neq 1$, minimality of Y gives $[Y,F] = Y$, and hence $Y \leq F$. If $[Y,F] = 1$ for every direct factor F, then $Y \leq Z(Q) = 1$, a contradiction. Hence Y is contained in one of the displayed direct factors. It follows that the nonabelian minimal normal subgroups of Q are precisely the simple direct factors of Q_0, and the abelian minimal normal subgroups of Q are precisely the subgroups V_a with $a \in I$.

Thus Q_0, being generated by all nonabelian minimal normal subgroups of Q, is characteristic in Q. Let $R = \prod_{a \in I} Q_a$. Since Q_0 is centerless, $R = C_Q(Q_0)$, so R is characteristic in Q.

Since the abelian minimal normal subgroups V_a have pairwise distinct orders, each V_a is characteristic in Q. We now recover each Jacobi quotient factor Q_a from these characteristic subgroups. For $a \in I$, let $W_a = \prod_{b \in I,\ b \neq a} V_b$, with the empty product allowed. Since R and all V_b are characteristic in Q, the subgroup $C_R(W_a)$ is characteristic in Q. In Q_a, one has $C_{Q_a}(V_a) = V_a$, because centralizing V_a forces the S-coordinate to act trivially on V_a, and the natural action is faithful. Hence

$$C_R(W_a) = Q_a \times \prod_{b \in I,\ b \neq a} V_b.$$

Taking commutator subgroups gives $C_R(W_a)' = Q_a$, since Q_a is perfect and the subgroups V_b are abelian. Thus each Q_a is characteristic in Q.

Therefore every automorphism of Q preserves Q_0 and every Q_a. The corresponding factors D_{2,x_2} and $J_{p,n(p,i)}$ are perfect, and their centerless quotients are respectively Q_0 and the Q_a's. By Lemma 2.6, the α-ratio of D_b is the product of the α-ratios of the displayed factors.

Using Proposition 3.7 and Corollary 3.10, we obtain

$$\alpha(D_b) = 2^{-x_2} \prod_{\substack{p \le P \\ p \text{ odd}}} \left(\frac{p-1}{p}\right)^{x_p}.$$

We compute valuations. For an odd prime $q \le P$, the denominator contributes $-x_q$, and the numerator factors $\ell - 1$ contribute to v_q only from primes $\ell > q$. Therefore

$$v_q(\alpha(D_b)) = -x_q + \sum_{\substack{\ell > q \\ \ell \le P \\ \ell \text{ prime}}} x_\ell v_q(\ell - 1) = -e_q$$

by the defining recursion. Similarly,

$$v_2(\alpha(D_b)) = -x_2 + \sum_{\substack{\ell > 2 \\ \ell \le P \\ \ell \text{ prime}}} x_\ell v_2(\ell - 1) = -e_2.$$

Finally, if $q > P$, then q does not occur in any denominator factor, and if $q \mid \ell - 1$ for some prime $\ell \le P$, then $q < \ell \le P$, a contradiction. Hence $v_q(\alpha(D_b)) = 0 = -e_q$ for every prime $q > P$. Thus $v_q(\alpha(D_b)) = -e_q$ for every prime q, and so $\alpha(D_b) = \prod_p p^{-e_p} = 1/b$. The group D_b is finite and perfect because it is a finite direct product of finite perfect groups. $\square$

3.5 The final construction

We now combine denominator blocks with prime-avoiding integer blocks.

Theorem 3.12. *For every positive rational number r, there exists a finite group G such that*

$$\frac{|\operatorname{Aut}(G)|}{|G|} = r.$$

Proof. Let $r = a/b$, with $a, b \in \mathbb{Z}^+$. Clearly, all values within $\mathbb{Q}^+$ may be represented in this form. If $b = 1$, then Lemma 3.1, applied with $d = a$ and $\Pi = \varnothing$, gives a finite group G with $\alpha(G) = a$.

Assume now that $b > 1$. By Proposition 3.11, choose a finite perfect group D_{2b} with $\alpha(D_{2b}) = 1/(2b)$. Let Π be the set of primes dividing $|D_{2b}|$. Since $|D_{2b}| = 2b|\operatorname{Aut}(D_{2b})|$, the order of D_{2b} is even, and hence $2 \in \Pi$. Apply Lemma 3.1 with $d = 2a$ and this set Π. The parity hypothesis in that lemma is satisfied, and we obtain a finite group A such that $\alpha(A) = 2a$ and no prime in Π divides $|A|$. Hence $\gcd(|A|, |D_{2b}|) = 1$.

By Lemma 2.7,

$$\alpha(A \times D_{2b}) = \alpha(A)\alpha(D_{2b}) = (2a)\frac{1}{2b} = \frac{a}{b}.$$

Thus $G = A \times D_{2b}$ has the required ratio. $\square$

4 Further research directions

The construction above realizes every positive rational value, but it is not optimized and is only partially explicit in a practical sense. We now discuss several directions for refinement and classification.

Question 4.1. For $r \in \mathbb{Q}^+$, let

$$m(r) = \min\{|G| : G \text{ is finite and } \alpha(G) = r\}.$$

Can one give a useful formula, sharp effective bounds, or asymptotic estimates for $m(r)$ in terms of r?

The theorem proves that $m(r)$ is finite for every $r \in \mathbb{Q}^+$, but the groups constructed here are usually very large. The dyadic construction is especially inefficient, since it uses sporadic simple groups and asymmetric binary subsets. Even for ratios such as $r = 1/p$, the present construction does not appear designed to be close to minimal.

Question 4.2. Can the dyadic denominator blocks be replaced by a uniform construction using quasisimple groups of Lie type, or universal central extensions of groups of Lie type, instead of sporadic simple groups?

The odd-prime construction is uniform and uses the groups $H(V) \rtimes \mathrm{Sp}(V)$. By contrast, the dyadic construction uses ATLAS data for the Baby Monster, the Rudvalis group, and $PSL_3(4)$, together with a finite-code argument. A uniform $p = 2$ analogue of the Jacobi block, or a uniform family of quasisimple dyadic denominator blocks, would make the proof more conceptual.

Question 4.3. For which natural classes $\mathcal{C}$ of finite groups is the set

$$\{\alpha(G) : G \in \mathcal{C}\}$$

equal to $\mathbb{Q}^+$?

The finite abelian case does not realize all positive rationals. As aforementioned, McCulloch proved that if $r = a/b$ is in lowest terms and $r = \alpha(A)$ for a finite abelian group A, then b is squarefree. He also proved that no odd prime occurs as such a value [McC26]. The finite nilpotent and finite solvable cases are especially natural next targets. The related work of Bray and Wilson studies the size of $\mathrm{Aut}(G)$ relative to $\varphi(|G|)$ in perfect and soluble groups, but those results concern a different exact-realization problem [BW05; BW06]. González-Sánchez and Jaikin-Zapirain constructed finite p-groups G_i such that $|\operatorname{Aut}(G_i)|/|G_i| \to 0$ [GJ15], but exact realization of specific rational values is a different problem.

Question 4.4. Can one classify the exact image of α on finite abelian groups, on finite p-groups for a fixed prime p, on the union of all finite p-groups, on finite nilpotent groups, or on finite solvable groups?

For finite abelian groups, McCulloch's results already give strong obstructions and examples, but they do not provide a complete description of the image [McC26]. In the nilpotent case, finite nilpotent groups decompose as direct products of their Sylow subgroups, and the Sylow factors have coprime orders. Thus the finite nilpotent image is controlled by the images arising from finite p-groups, together with coprime-order multiplicativity. A classification in nilpotent or solvable classes would clarify how much nonabelian structure is needed to realize arbitrary rational values.

Question 4.5. Given $r \in \mathbb{Q}^+$, what can be said about the finite groups G satisfying $\alpha(G) = r$?

The proof constructs one group for each rational value, but it does not attempt to describe all groups with a given value. The identity $\alpha(G) = |\operatorname{Out}(G)|/|Z(G)|$ suggests that the fibers of α should reflect the interaction between central structure, outer automorphisms, and the quotient $G/Z(G)$.

Question 4.6. Can one give an efficient and implementable algorithm which, given $r = a/b \in \mathbb{Q}^+$, outputs a finite presentation, permutation representation, or matrix-group description of a finite group G with $\alpha(G) = r$, together with a verifiable certificate of the ratio?

The odd-prime Jacobi blocks have compact matrix-group descriptions, and the metacyclic numerator blocks are elementary. The main computational obstacle is the production of practical dyadic blocks. Replacing the sporadic-group input by smaller uniform groups, or giving explicit asymmetric binary subsets in all required dimensions, would make the construction substantially more accessible to implementation in systems such as GAP or Magma.

Question 4.7. Are there other natural families of finite groups G for which $|Z(G)|$ and $|\operatorname{Out}(G)|$ can be controlled exactly enough to realize prescribed denominator factors?

The Jacobi blocks give $\alpha(J_{p,n}) = (p-1)/p$ for odd primes p. The factor $p-1$ is harmless for the recursive construction, but a direct mechanism producing ratios such as $1/p$ or p^{-e} for odd p would simplify the denominator argument. More generally, it would be useful to understand when semidirect products with extraspecial or class-two p-groups yield exactly computable outer automorphism groups.

A Verification script for the explicit binary subsets

The following Python 3 script recomputes the additive-triple refinement used in Lemma 3.6. Addition in $\mathbb{F}_2^e$ is implemented by bitwise xor. The script verifies that the computed refinement tables are exactly the tables displayed in Lemma 3.6, that the terminal partitions are discrete, and that the two sets contain the standard bases of their ambient vector spaces.

```
from collections import Counter, defaultdict

S5 = [1, 2, 4, 8, 16, 17, 21, 24, 26, 27, 30, 31]
S6 = [1, 2, 4, 6, 7, 8, 16, 24, 26, 32, 40, 44, 47]

E5 = [
    ((1, 4, 26), (2, 8, 21), (16, 17, 24, 27, 30, 31)),
    ((1,), (2, 21), (4,), (8,), (16,), (17,), (24,), (26,), (27, 30), (31,)),
    ((1,), (2,), (4,), (8,), (16,), (17,), (21,), (24,), (26,), (27,), (30,), (31,))
]

E6 = [
    ((1, 16, 26, 32, 44, 47), (2, 4, 6, 7, 8, 24), (40,)),
    ((1, 16, 26), (2, 6), (4,), (7, 8), (24,), (32, 44, 47), (40,)),
```

```
    ((1,), (2,), (4,), (6,), (7,), (8,), (16,), (24,), (26,), (32, 47), (40,), (44,)),
   ((1,), (2,), (4,), (6,), (7,), (8,), (16,), (24,), (26,), (32,), (40,), (44,), (47,))
]

def blocks(colors):
    out = defaultdict(list)
    for x, c in colors.items():
        out[c].append(x)
    return tuple(sorted(tuple(sorted(v)) for v in out.values()))

def refine(S):
    T = set(S)
    colors = {x: sum((x ^ y) in T for y in S) for x in S}
    out = [blocks(colors)]
    while True:
        data = {}
        for x in S:
            pairs = Counter((colors[y], colors[x ^ y]) for y in S if (x ^ y) in T)
            data[x] = (colors[x], tuple(sorted(pairs.items())))
        labels = {v: i for i, v in enumerate(sorted(set(data.values()), key=repr))}
        colors = {x: labels[data[x]] for x in S}
        part = blocks(colors)
        if part == out[-1]:
            return out
        out.append(part)

def check(S, E, e):
    ok = len(S) == len(set(S))
    ok = ok and all(0 < x < 2**e for x in S)
    ok = ok and set(2**i for i in range(e)) <= set(S)
    ok = ok and refine(S) == E
    ok = ok and all(len(b) == 1 for b in E[-1])
    if not ok:
        raise SystemExit(f"S_{e} failed")

check(S5, E5, 5)
check(S6, E6, 6)
print("Lemma 3.6 certificates verified.")
```

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